

GLOBAL ROBOTICS CHALLENGE ROV TECHNICAL REPORT

M & P ROBOTIX

Remotely Operated Vehicle — Technical Documentation



Team Name	M & P Robotix
Competition	Global Robotics Challenge (GRC)
Category	ROV – Rangers Level
Report Year	2026

Team Members

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Coach: Christine Youssef Fayez



مركز القديسين للتنمية
KEDESEEN DEVELOPMENT CENTER

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About This Report

This document is the official technical report for the M & P Robotix ROV entry in the Global Robotics Challenge. It covers the complete design and build process from initial concept through final pool testing. Each section documents a specific area of the project including mechanical design electrical systems software safety and competition strategy. The goal is to clearly explain every decision the team made and why it was the right choice for the requirements of this competition.

Competition: Global Robotics Challenge **Category:** ROV – Rangers Level

Abstract

The world's oceans are under increasing pressure from pollution, climate change, and unsustainable energy practices. These challenges affect marine ecosystems on a global scale and demand practical, technology-driven solutions. The Global Robotics Challenge provided our team with a platform to address some of these challenges through the design and deployment of an underwater remotely operated vehicle.

To fulfill the mission requirements, the vehicle needed to be compact, highly maneuverable, and mechanically reliable under the pressure and conditions of underwater operation. The final design incorporates six thrusters for full six-degree movement, two independent manipulator systems for object interaction, and 2 high-definition cameras for piloting and computer vision tasks. Cost efficiency was a priority throughout the build, achieved by fabricating key components in-house and selectively reusing tested parts from previous builds.

A team of eighteen students, each assigned to a specific engineering role, contributed to the design, fabrication, and testing of the vehicle. This report documents every major decision made during that process.

6

Thrusters

1

Manipulators

2

HD Cameras

1 Team Members

The M & P Robotix team consists of eighteen students working under the guidance of Coach Christine Youssef Fayez. Each member was assigned a specific role based on their skills and interests covering mechanical design electrical systems software development and fabrication.

Full Name	Date of Birth	Role
Parthinya Tamer	17 Aug 2008	Mechanical
Mohraeel Atef AbdEl-Malak	27 Sep 2008	Electrical Hardware
Mark Youssef Mikhail Awada	26 Mar 2009	Electrical Software
George Adel Labib	13 Jun 2009	Electrical Software
Fady George Samy	16 Aug 2008	Mechanical
Clara Youssef Fayez	08 Oct 2009	Electrical Hardware
Joya Sameh Nagy	02 Nov 2009	Mechanical
Jolie John Joseph	21 Jul 2008	Electrical Software
Marina Morcos Samir	25 Sep 2008	Mechanical
Pelajia Emad Fayez	16 Nov 2008	Mechanical
Pavly Ashraf	22 Jul 2008	Electrical Hardware
Veronica Ramzy Sobhy	11 Nov 2007	Electrical Software
Mariam Maged Asaad	10 Mar 2009	Mechanical
Claire Ibrahim	15 Feb 2008	Electrical Software
Kathreen Gamal Zakaria	03 Apr 2008	Electrical Software
Mayven Tarek Halim	07 Jul 2008	Electrical Software
Mina Wagih Kamal	18 Mar 2009	Electrical Software

Coach: Christine Youssef Fayez

2 Design Rationale & Evolution

The design of our ROV is the result of a structured development process that drew on lessons from previous builds and the specific technical requirements of this year's competition missions. Rather than designing from scratch, the team chose to evolve an existing design, refining areas that performed poorly and preserving what worked well.

Three Phase Design Process

Phase 1 — Brainstorming

All team members participated in an open brainstorming session before any CAD tools were opened. Every idea was recorded and discussed regardless of feasibility. This ensured that no potential solution was dismissed too early and that all engineering disciplines contributed to the initial direction.

Phase 2 — Decision Making

The mechanical team developed several candidate designs and documented the trade-offs of each. These options were then reviewed by the electrical team to assess compatibility with the electronics layout. After cross-discipline review, a final design was selected by consensus and approved before fabrication began.

Phase 3 — Design & Simulation

The approved design was modeled in SolidWorks. ANSYS was used to run Computational Fluid Dynamics simulations to verify drag coefficients, lift forces, and thruster positioning. Fabrication files including STL and DXF formats were exported from SolidWorks for CNC routing, laser cutting, lathe machining, and 3D printing.

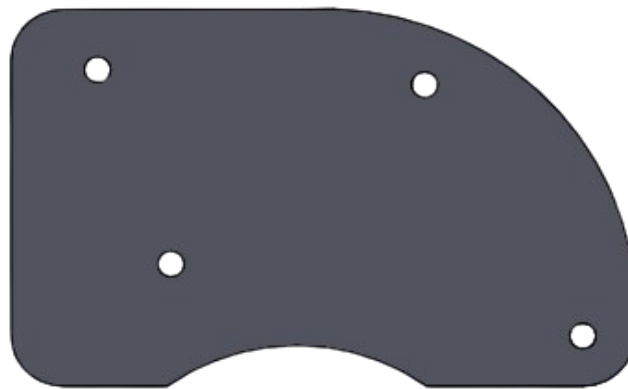


Figure 1 — SolidWorks CAD component from the structural design phase

3 Mechanical Structure

The mechanical structure defines the physical shape, weight distribution, and rigidity of the ROV. Every design decision in this section was evaluated against three criteria: weight reduction, drag minimization, and ease of assembly for maintenance and competition preparation.

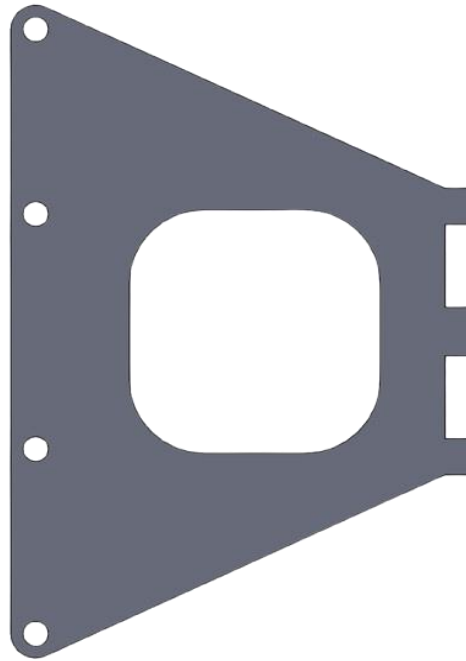


Figure 2 — Vertical plate component showing thruster mounting ports

Material Selection

Polyurethane was selected as the primary structural material due to its favorable combination of low density, mechanical strength, and low cost. Parts were fabricated at local workshops using laser cutting equipment and lathe machines. This approach reduced manufacturing costs significantly. Acrylic was used for the control housing dome cover due to its transparency which allows visual inspection of internal components. Aluminum alloy was used for camera enclosures to provide structural rigidity against water pressure while keeping mass low.



Figure 3 — Polyurethane structural plate showing geometry and bolt pattern

Control Housing

The control housing is a sealed enclosure that contains all electronic components including circuit boards and the microcontroller. The integrity of this enclosure is critical because water intrusion would render the entire electrical system inoperable. The sealing system consists of five layers applied in sequence: two O-rings seated in precision-machined grooves, an acrylic circular backing plate, a rubber gasket layer, an acrylic dome-shaped outer cover, and an aluminum compression ring that distributes clamping force evenly across the dome to prevent acrylic deflection. All layers are secured with fasteners to maintain consistent compression.

The dome profile was adopted from designs used by higher-class competition teams and has a demonstrated advantage in reducing drag by allowing water to flow smoothly around the housing rather than impacting a flat surface.

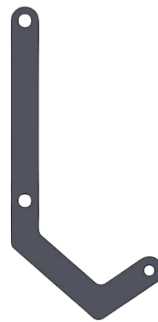


Figure 4 — Angular bracket component used to connect frame elements

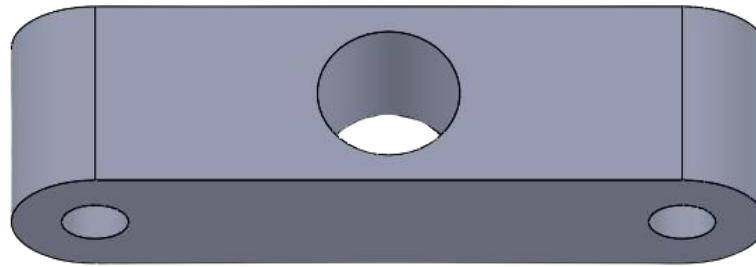


Figure 5 — Plate component with drag-reduction slots



Figure 6 — Extended structural plate showing full geometry

4 Propulsion System

The propulsion system determines the ROV's range of motion, speed, and precision of movement underwater. The configuration was designed to provide full six-degree-of-freedom control while remaining within the team's budget and power supply limitations.

Thruster Configuration

The ROV uses a total of six thrusters. Two Bilges Blue Robotics thrusters are mounted vertically on the upper plate to control heave (vertical movement) and pitch (nose angle). The remaining four thrusters are modified bilge pumps mounted on the vertical connector plates at angles ranging from 30 to 60 degrees. This angular configuration provides effective forward, backward, and lateral thrust simultaneously.



Figure 7 — 3D model of the modified bilge pump thruster mount

Bilge Pump Modification

Deploying six commercial Bilge thrusters was not feasible within the team's budget and would have exceeded the available current supply. As an engineering solution, the team modified standard bilge pumps by replacing the internal impellers with Water Pump Bilge thruster propellers and fitting externally fabricated kort nozzles. The adapter components required to mount the propeller and nozzle assembly were designed in SolidWorks and produced using the team's 3D printer. The kort nozzle concentrates water flow through the propeller disc, increasing thrust output relative to an open propeller configuration. This modification produced significantly better performance than the unmodified bilge pump at a fraction of the cost of additional Bilge units.

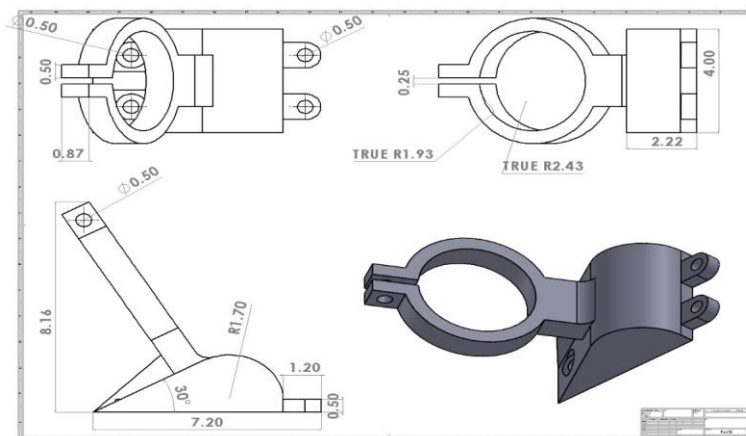


Figure 8 — Engineering drawing of bilge pump adapter (30 degree mount angle)

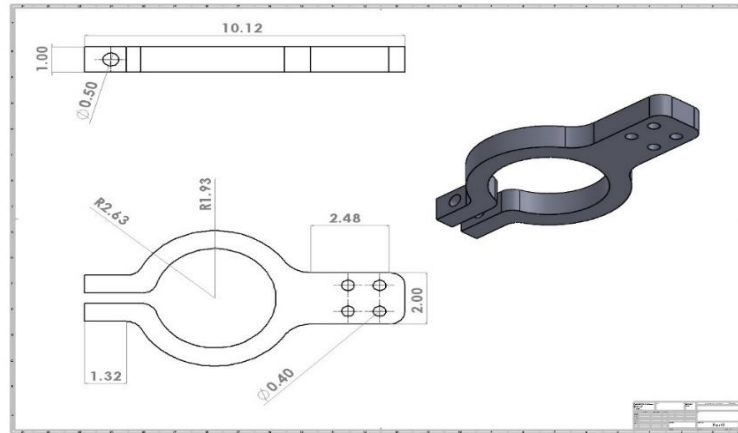


Figure 9 — Korte nozzle housing with fabrication dimensions

5 Sealing & Waterproofing

Waterproofing is one of the most operationally critical aspects of ROV design. Even minimal water ingress into the electronics enclosure can cause irreversible component failure. The team implemented consistent sealing solutions across all subsystems that required protection from the aquatic environment.

Electric Cable Penetrations

All electrical cables that pass through the wall of the control housing are routed through pneumatic threaded fittings. Each cable is first inserted into a flexible pneumatic hose, and the hose is then secured to the fitting. Teflon tape is applied to all fitting threads prior to installation to ensure a pressure-tight connection. This method provides a reliable seal without the need for adhesives or epoxy that could complicate future maintenance.

Camera Enclosures

Custom aluminum housings were fabricated for the two manipulator view cameras using a lathe machine. Each housing is a two-piece assembly with male and female sections joined by a threaded interface. O-rings are seated in machined grooves at the mating face to provide the primary fluid seal. Cable exit is managed through a pneumatic fitting at the rear of the housing, connecting via hose to the main control enclosure. All housings were pressure tested using compressed air prior to camera installation.

DC Motor Shaft Seals

Rotating shaft penetrations present a particular sealing challenge since the seal must prevent fluid ingress while allowing continuous rotation. This is resolved using a mechanical shaft seal and dual bearings on the output side of each motor. On the opposite side, two O-rings are compressed between

an acrylic endplate and the polyurethane motor body, and a pneumatic fitting handles the cable exit to the control housing.



Figure 10 — Assembled sealed thruster housing

Test Protocol

Every sealed assembly was validated through two tests before electronic components were installed. In the pressure test, the enclosure was connected to a compressed air source and pressurized to verify integrity. In the pool immersion test, the complete ROV was submerged for a minimum of ten minutes with no electronics installed and then inspected thoroughly for any moisture entry.

6 Buoyancy

The ROV was designed to be positively buoyant, meaning it naturally tends to rise when submerged. This design choice reduces the continuous load on the vertical thrusters during depth-hold operations and provides a passive failsafe: if the vehicle loses power during operation it will rise to the surface independently rather than descending to the pool floor.

Calculation Method

Component densities, volumes, and masses were catalogued in a SolidWorks model. The total buoyancy force was calculated by summing the displaced water volume across all components and applying Archimedes' principle. The calculated total buoyancy force is 125.939 Newtons. Following physical construction, the calculated values were compared against measured data and foam was added where required to achieve the target buoyancy margin.

Component Buoyancy Data

Component	Material	Density kg/m ³	Volume m ³	Buoyancy N
Control Housing	PA Type 6	1120	0.001131	41.845
Aluminum Ring	1060 Alloy	2700	0.000118	1.155
Housing Locker x4	PA Type 6	1120	0.000417	1.023
Lower Plate	PA Type 6	1120	0.000518	5.080
Pneumatic Manipulator	Various	NA	0.000794	22.039
Kort Nozzle & Bilge x4	Various	NA	0.001273	3.985
Upper Plate x2	PA Type 6	1120	0.000353	1.732
Dome	Acrylic	1200	0.000522	5.119
Camera Housing x2	1060 Alloy	2700	0.000244	2.555
Thruster x2	Various	NA	0.000390	1.989

Achieving the correct buoyancy balance meant the pilot could maintain stable depth with minimal vertical thruster input, significantly improving operational precision during mission tasks.

7 Electrical Systems

Dual Board Architecture

The electrical system is organized across two separate printed circuit boards. The first board manages power regulation and communication signals. The second board is dedicated to motor drivers. This separation isolates high-current motor circuits from the sensitive signal-processing components, simplifying diagnostics and reducing the risk of signal interference. All connectors are modular, enabling individual components such as sensors, motors, and cameras to be removed and replaced without soldering.

Power System

The ROV operates from a 12-volt DC supply delivered through the tether. A buck converter on the signal board steps this down to 5 volts for components requiring lower operating voltage. The tether uses 11 AWG silicone insulated power cables selected for their low mass, flexibility, and current capacity. Voltage drop across the full tether length is approximately 10 percent, which is compensated by onboard filter capacitors to maintain stable supply to all components.

Sensors

An IMU sensor (MPU6050 / MPU9250) provides continuous measurement of the vehicle's angular rate and heading. This data feeds the onboard stability controller. A raindrop sensor inside the housing monitors for humidity, providing an immediate alert in the event of water ingress before damage escalates.

Motor Control Board

The motor board houses 3 Cytrons motor drivers rated at 18 amps each and one MC33886 motor driver rated at 6 amps, alongside Electronic Speed Controllers for the brushless thrusters. Motor drivers amplify the low-current PWM control signals from the Arduino into the high-current outputs required to drive the thruster motors.

Tether

The tether is 22 meters in length, calculated using the Pythagorean theorem to ensure diagonal pool coverage (15 meters) with a 6-meter surface reserve for connection to the control station. It carries 12V power, CAT6e ethernet for camera feeds and communication, and a polyurethane pneumatic hose for compressed air to the pneumatic manipulator. All conductors are bundled within a flexible, abrasion-resistant outer sheath that protects against interlacing and mechanical damage.

8 Software & Communication

Communication Architecture

Communication between the surface control station and the ROV is implemented over UDP (User Datagram Protocol). UDP was selected for its minimal transmission overhead and low latency characteristics, both of which are essential in a real-time vehicle control context. Control data is encoded

using Google Protocol Buffers, reducing packet size from 135 bytes (JSON equivalent) to 31 bytes per message. This results in faster throughput and lower CPU utilization on both ends of the link.

Onboard Controller

An Arduino Mega serves as the onboard hardware controller. It receives UDP packets via an ENC28J60 ethernet-to-SPI module, parses incoming joystick axis values, and generates PWM outputs for each motor driver in a continuous control loop. This architecture ensures minimal latency between operator input and mechanical response.

Active Stability System

A PID (Proportional–Integral–Derivative) controller is implemented in firmware to provide active heading and attitude stabilization. The IMU sensor provides angular data which is processed through complementary filters to produce accurate heading estimates. The PID controller compares the current heading against the pilot's set-point and applies corrective thruster outputs automatically. This allows the pilot to focus on mission tasks rather than continuously compensating for environmental disturbances such as water currents.

Graphical User Interface

The topside GUI is built using Electron, a cross-platform desktop application framework. The interface displays live video feeds from the 2 cameras, provides controls for maximum thruster speed and PID gain parameters, and includes a measurement overlay for the fish-size estimation task. UDP socket communication is handled through Chrome extension APIs embedded within the Electron runtime.

9 Computer Vision & Image Processing

Image processing is performed on the surface computer, which has the computational resources required for real-time analysis. The onboard Arduino does not have sufficient processing capacity for vision tasks. OpenCV is used as the primary processing library.

Photomosaic Construction

The photomosaic mission requires assembling eight sequentially captured images into a single composite image. The pipeline applies image enhancement filters, initializes an ORB (Oriented FAST and Rotated BRIEF) feature detector, extracts keypoints and descriptors from each frame, and uses a Brute Force Matcher to identify corresponding features between adjacent images. Matched images are progressively stitched together until all eight frames form a single seamless mosaic.

10 Payload Tools

The payload system enables the ROV to physically interact with objects and complete manipulation tasks during competition missions. Two distinct manipulator designs are employed simultaneously to maximize operational efficiency.

Pneumatic Manipulator

The second manipulator is actuated by a pneumatic piston driven by compressed air supplied through the tether. Pneumatic actuation provides fast response and high grip force with simple binary control requiring a single operator input. The end effector is a flat paddle design with rubber contact pads for friction, suited to handling thin or planar objects. An integrated DC motor and gear set enable rotational positioning of the end effector, expanding the range of objects that can be retrieved without repositioning the ROV.

Camera System

Three Hikvision HD analog cameras are deployed on the ROV. Analog cameras were specified over digital units due to their negligible signal latency, which is critical for responsive real-time piloting. Two cameras are mounted at an angle above the manipulators on articulated rotating brackets, allowing the pilot to adjust viewing angle on both axes for precise grip alignment.

11 Safety

Safety is a fundamental requirement at every stage of the project, from initial design through workshop fabrication, testing, and competition deployment. The team maintains a structured safety program that governs how members work, how the ROV is built, and how it is operated at the competition venue.

Safety Philosophy

A safe working environment protects team members from injury, protects equipment from damage, and improves overall productivity by reducing time lost to accidents and repairs. The team's safety standards are aligned with GRC competition requirements and draw from professional industry practices documented in the Oceaneering Americas Region HSE Employee Handbook.

Workshop Protocols

Before any high-risk fabrication procedure is started, a Job Safety Analysis form is completed. This form requires team members to identify every hazard associated with the task and document mitigation measures before work begins. Safety checklists are reviewed at the start of each workshop session. When new equipment or processes are introduced, new JSA forms are written and approved before use.

Training is provided to all team members covering electrical safety, tool handling, back and posture safety, hazardous materials handling, and workshop housekeeping procedures. New members are supervised directly by experienced members during their first sessions with any equipment and are only cleared to work independently after demonstrating correct and safe technique. Monthly safety drills are conducted to ensure all members can respond effectively to emergency scenarios.

ROV Vehicle Safety Features

The following safety features are incorporated into the vehicle design and verified during pre-operation inspection:

- O-ring face seals and epoxy potting on all electrical penetrations prevent water ingress and associated electrical hazard
- Cap nuts are applied to all exposed screw ends and the frame design eliminates sharp external edges to prevent lacerations during handling
- Propeller guards are installed on both faces of every thruster. Guard apertures do not permit passage of a circle greater than 8 to 10 mm in diameter in compliance with GRC safety requirements
- Tether strain reliefs at both the ROV and surface ends prevent mechanical stress from reaching internal electrical connections
- Independent 12VDC and 220VAC emergency cutoff switches are accessible on the surface control box and can immediately de-energize the entire system
- Humidity, temperature, and pressure sensors inside the control housing transmit telemetry to the pilot GUI, providing real-time alerts for leakage or thermal events
- A 25-ampere fuse is installed within 30 cm of the power supply output to protect all downstream wiring from overcurrent events
- Hazard warning labels are affixed to all high-voltage components and moving parts on the ROV

Pre-Operation Safety Checklist

All team members follow a standardized safety checklist before, during, and after every deployment. Inspections are carried out at the workshop prior to transport and repeated at the competition pool deck before the ROV enters the water. The checklist covers structural integrity, electrical continuity, seal condition, tether routing, and emergency cutoff switch functionality.